R18

Code No: 157AM

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What is well defined problem?	[2]
b)	Define search tree and write its properties.	[3]
c)	What is game tree?	[2]
d)	Explain evaluation function.	[3]
e)	Define atomic and complex sentence in first order logic.	[2]
f)	What is First order logic?	[3]
g)	Define temporal constraints.	[2]
h)	Explain classical planning.	[3]
i)	Define over fitting.	[2]
j)	State Baye's rule.	[3]

PART – B

(50 Marks)

2.a)	What is simple problem solving agent? Explain it briefly.	
b)	Discuss Greedy best first search algorithm.	[5+5]
	OR	
3.a)	What is bidirectional search ? Explain in detail.	
b)	Explain Breadth First Search algorithm with an example.	[5+5]
4.a)	Elaborate on knowledge based agents.	
b)	Explain CSP problem for job scheduling.	[5+5]
	OR	
5.a)	What is Resolution? Explain Resolution algorithm for Proposition logic.	
b)	Explain backtracking searching technique.	[5+5]
6.a)	Explain quantifier estimates and its types.	
b)	Analyze the steps in knowledge-engineering process.	[5+5]
	OR	
7.a)	Explain reasoning with default information.	
b)	Write simple forward chaining algorithm.	[5+5]

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8.a) Discuss regressive relevant state space search.
b) What is Graph Plan? Explain in detail. [5+5]

OR

9.a) Describe multi agent planning.
b) Give the solution for scheduling planning. [5+5]

10.a) Explain inductive logic programming.
b) Describe Bayesian networks in detail. [5+5]

OR

11.a) What is Decision tree? Explain steps to construct to Decision tree.
b) Explain the issues in decision tree learning. [5+5]

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